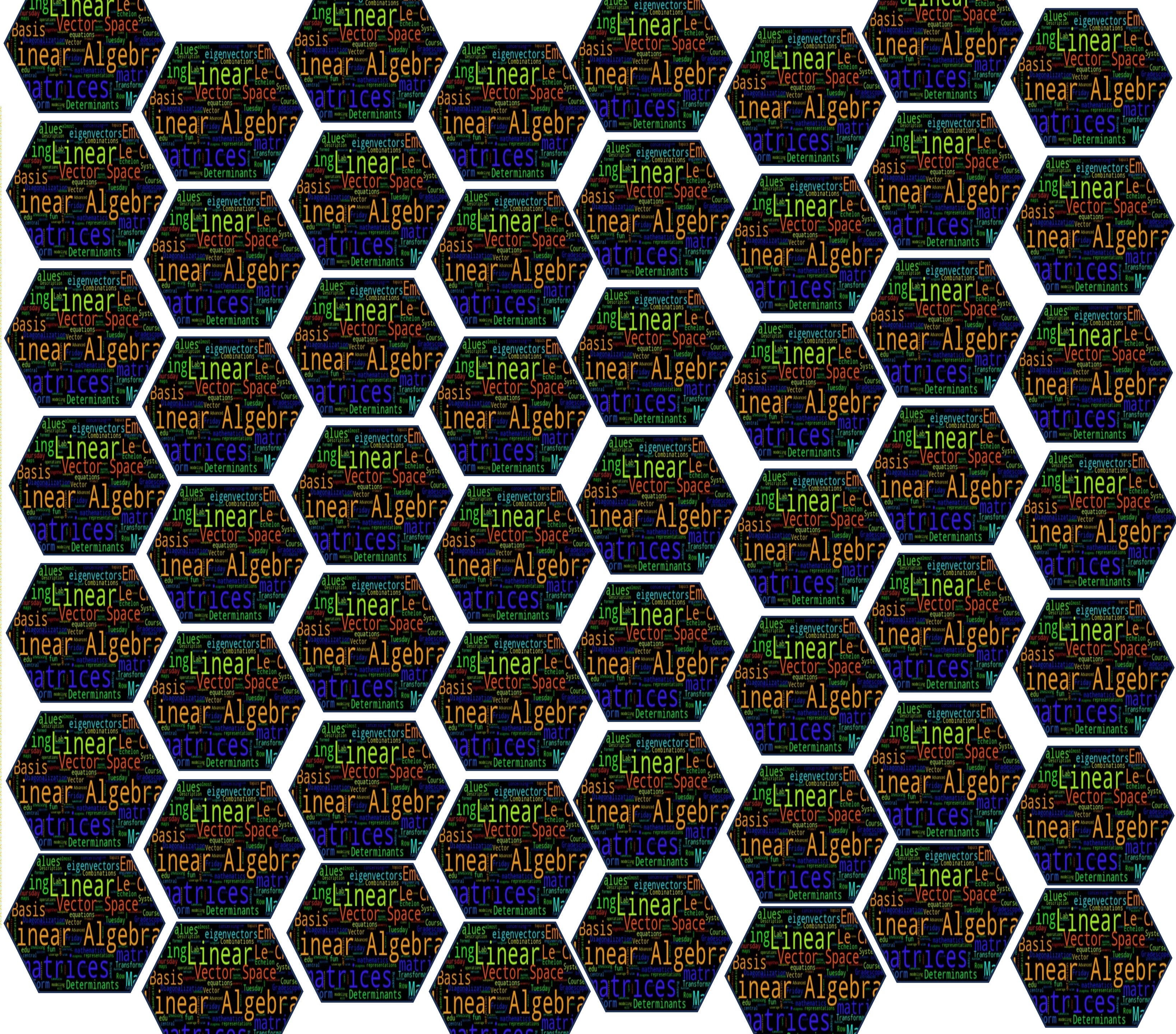
Linear Algebra



Lecture Contents

- Definition and Introduction
- Applications
- Topics in Linear Algebra
- Linear Equation and system of Linear Equations
- Solution Types

Scenario: The 3D Game Developer's Challenge

Imagine you've just landed your dream job at a top gaming company like **Ubisoft or Rockstar** Games. Your first task? To develop a new 3D adventure game where players explore a futuristic city. But there's a problem...**your objects aren't moving, rotating, or scaling correctly!** Buildings are floating, cars are upside down, and the game world is a mess!

Scenario: The 3D Game Developer’s Challenge

Your Mission:

- How can you **rotate** a game character correctly in 3D space? 🔄
- How do you **scale** objects to make them look realistic? 📏
- How can you apply **shadows, lighting, and physics**? ☀️

"The secret behind all of this? Linear Algebra! It’s the math that makes every modern game look smooth, realistic, and immersive."

Reveal the Truth:

- These transformations are done using **matrices**
- Moving characters? **Vector operations** ➡️
- Rotating a player's view? **Rotation matrices** 🔄
- Scaling objects? **Scaling matrices** 🔍

"Linear Algebra isn't just numbers—it's what makes the digital world move! Master it, and you can build the next-gen AI, animations, and immersive tech!"

Linear Algebra: Definition

- ❖ **"Linear"** → Because it deals with **straight-line relationships** (like scaling, adding, and transforming data in a predictable way).
- 🔹 **"Algebra"** → Because it involves **solving equations, manipulating symbols, and performing operations** on numbers, vectors, and matrices.
- ➢ **In short:** Linear Algebra studies **how data and geometric objects change in a structured, straight-line manner**, which is essential in fields like AI, graphics, and data science.

Definition and Applications

Linear Algebra is the mathematics of data, dealing with numbers, shapes, and patterns organized in grids (**matrices**) and lists (**vectors**).

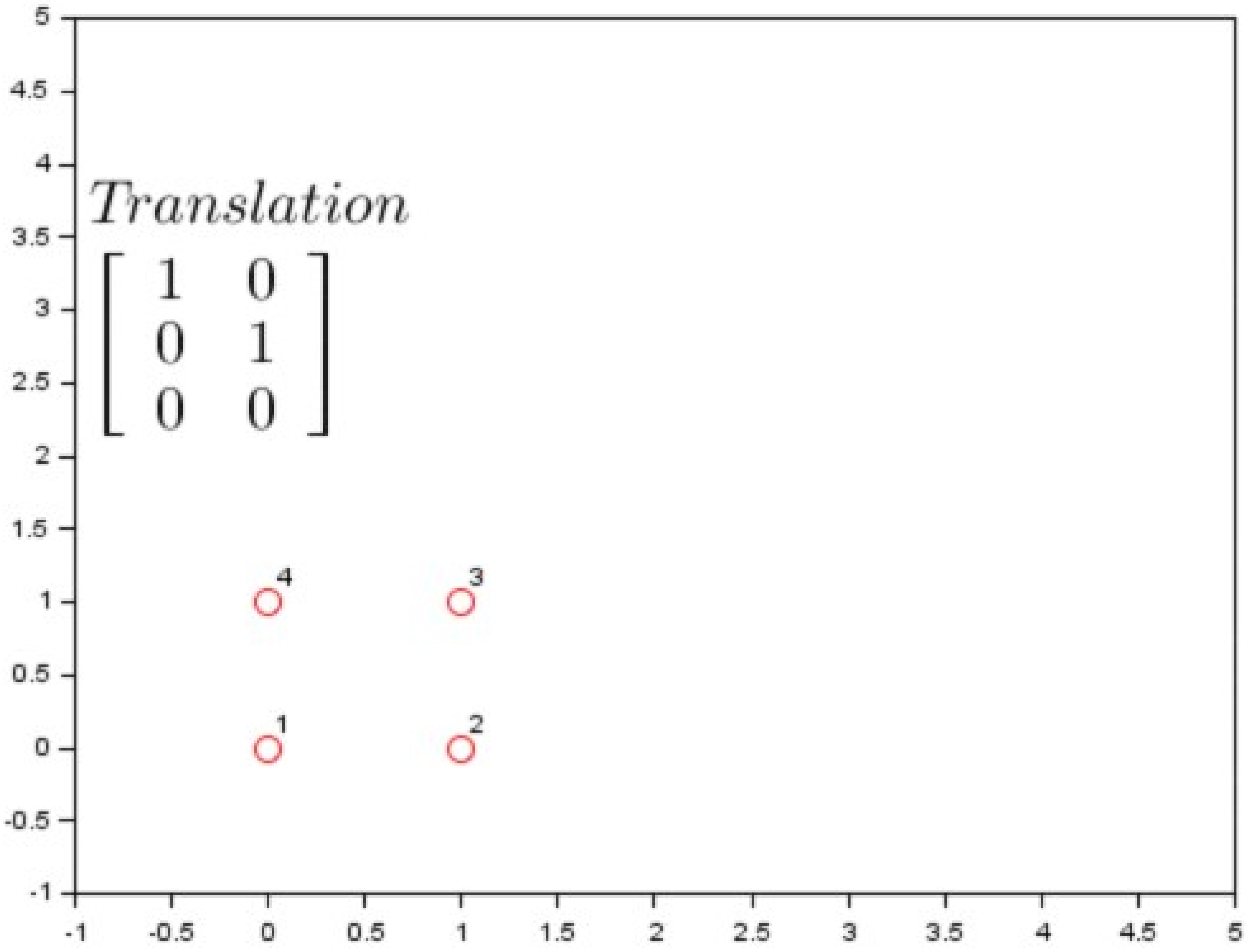
It helps in solving equations, transforming images, analyzing relationships in data, and optimizing algorithms.

Applications Of Linear Algebra

- Data Science
- Machine Learning
- Computer Graphics
- Cryptography
- Physics

and many more...

Transformation Matrix



How a Matrix is Built & Its Real-World Implementation

Imagine you're developing a

Netflix-like movie recommendation system.

You need to store user ratings for different movies.

You start with a **table of ratings** where:

Rows = Users (U1, U2, U3, ...)

Columns = Movies (M1, M2, M3, ...)

Entries = Ratings (1-5 stars)

How a Matrix is Built & Its Real-World Implementation

	M1	M2	M3	M4
U1	5	3	4	2
U2	4	2	5	3
U3	3	4	2	5

This table is actually a **matrix**!

$$A = \begin{bmatrix} 5 & 3 & 4 & 2 \\ 4 & 2 & 5 & 3 \\ 3 & 4 & 2 & 5 \end{bmatrix}$$

Now, we can use this matrix to find similar users and recommend movies using **Linear Algebra**.

How a Matrix is Built & Its Real-World Implementation

Finding Similar Users or Movies (Collaborative Filtering)

Goal: Find users or movies that are similar based on their rating patterns.

User-Based Filtering:

- Find users with similar tastes (e.g., if U1 and U2 gave similar ratings, they are similar).
- If U1 liked M3, and U2 hasn't watched it yet, recommend M3 to U2.

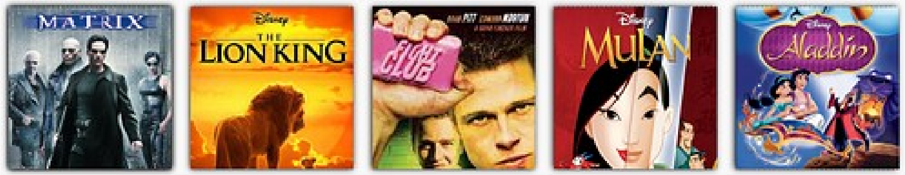
Movie-Based Filtering:

- If two movies (M1 and M3) get similar ratings from many users, they are similar.
- If U1 watched M1 and hasn't seen M3, suggest M3 to U1.

Rating Matrix r_{ui}

Explicit Feedback

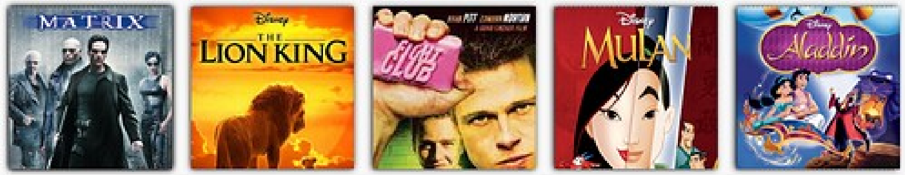
r_{ui} = *star rating 1 to 5*

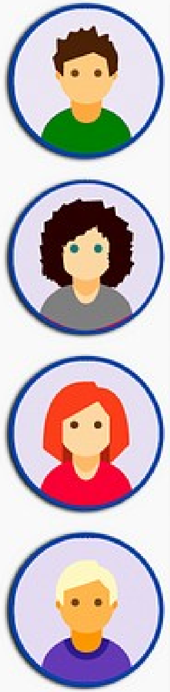


?	?	4	?	1
4	?	?	?	?
?	?	?	3	2
1	?	?	?	?

Implicit Feedback

r_{ui} = *did user watch the movie?*





1	0	1	0	1
1	1	0	1	1
0	1	0	1	1
1	0	0	1	0

Summary: How Netflix Uses Linear Algebra

1️⃣ **Matrix Representation:** Store user-movie ratings as a **matrix**.

2️⃣ **Similarity Computation:** Find similar users or movies (collaborative filtering).

3️⃣ **Matrix Factorization (SVD):** Predict missing ratings.

4️⃣ **Deep Learning (Autoencoders):** Learn complex user preferences.

5️⃣ **Continuous Learning:** Improve recommendations over time.

How Linear Algebra Helps in AI, Graphics, and Data Science

Situation: You Are Developing a Self-Driving Car System 🚗

Imagine you're working on **AI for a self-driving car**, and it needs to:

1️⃣ **Detect objects** on the road (pedestrians, vehicles, signs).

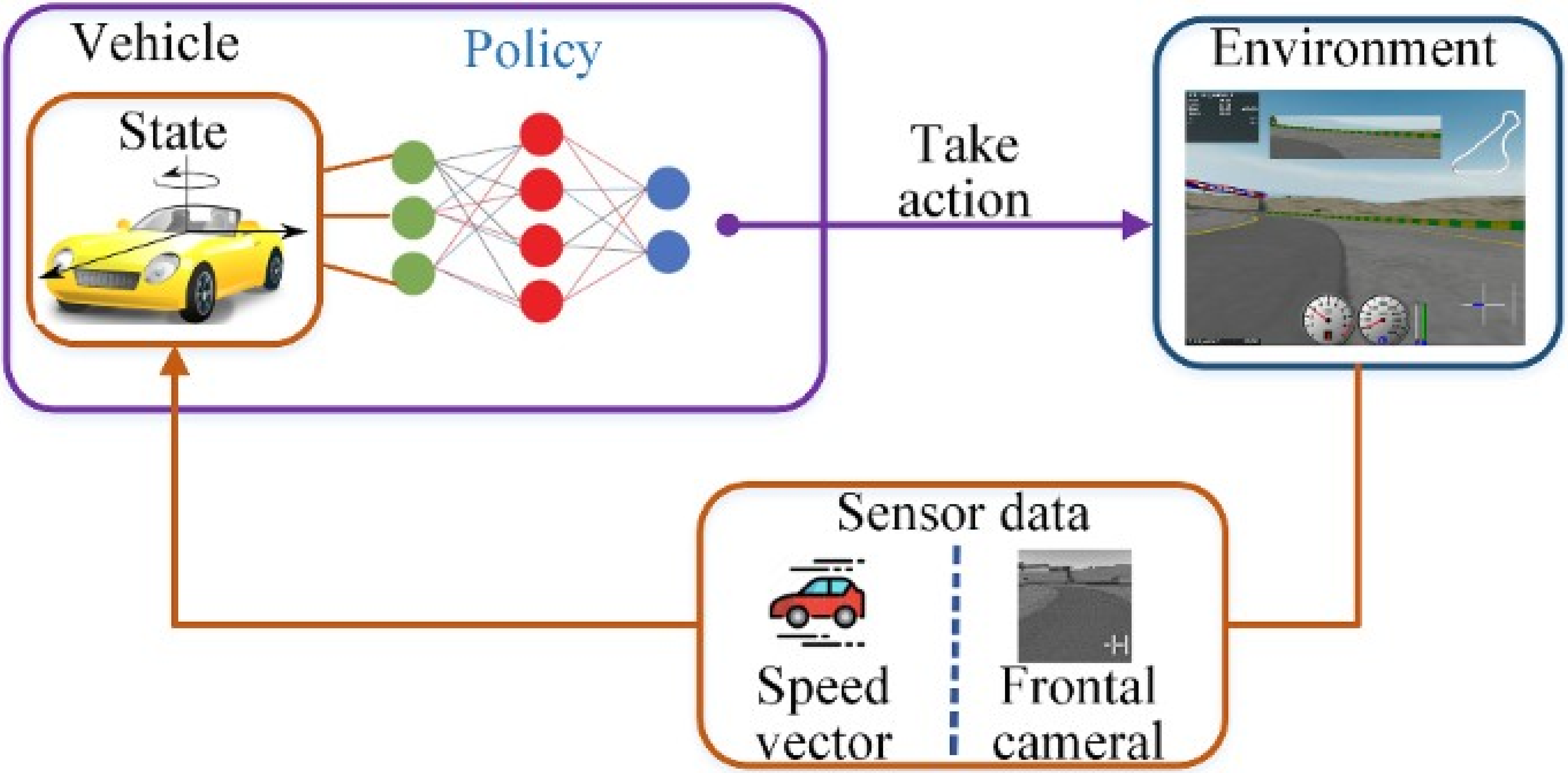
2️⃣ **Process sensor data** from cameras and LiDAR (Light Detection and Ranging).

3️⃣ **Navigate safely** by predicting movement.

How Does Linear Algebra Work Here?

- **1. Image as a Matrix (Graphics & AI)**
  - A grayscale image is just a **matrix of pixel values**.
  - Image transformations (blurring, edge detection) use **matrix operations**.
- **2. Object Detection Using Vectors (AI & Deep Learning)**
  - Every detected object is represented as a **vector**.
  - AI models classify objects by analyzing these vectors.
- **3. Predicting Car Movements (Data Science & Machine Learning)**
  - Car positions over time are stored as **matrices**.
  - **Linear regression** (a linear algebra technique) predicts future positions.

Self-Driving Car System 🚗



Some Topics of Linear Algebra

- System of Linear Equations
- Matrices
- Gaussian Elimination and Gaussian Jordan Method
- Vector Spaces
- Matrix Transformation
- Cryptography
- Eigen Values and Eigen Vectors and its applications to Machine Learning.

Linear Equation

An equation whose **exponent** or power is one is called linear equation.

1. $2x + 1 = 0$ → One Variable
2. $x + 3y = 7$ → Two Variables
3. $\frac{1}{2}x - y + 3z = -1$ → Three Variables
4. $x_1 - 2x_2 - 3x_3 + x_4 = \ln 2$ → Four Variables
5. $x_1 + x_2 + x_3 + \cdots + x_n = c$ → n Variables

- **Linear equation does not involve any products or roots of variables.**
- **All variables occur only to the first power and do not Appear as**
- **arguments of trigonometric, logarithmic, or exponential functions.**

Question: The following equations are linear or nonlinear? If it is nonlinear which term in each equation is making it nonlinear?

No

$$x + 3y^2 + 2z = 4$$

No

$$3x + 3y - xy = 5$$

No

$$sin(x) + 2y = 0$$

No

$$\sqrt{x_1} + x_2 + x_3 = 1$$

$$3x + 2y = cos\,20$$

Geometric Representation of Linear Equations

Variables	Equation	Geometric Representation
(nested table below)	(nested table below)	Straight Line in 2D space.
(nested table below)	(nested table below)	Plane in 3D space.
(nested table below)	(nested table below)	Hyperplane

Variables	Equation	Geometric Representation
$2, (x, y)$	$ax + by = c$	Straight Line in 2D space.
$3, (x, y, z)$	$ax + by + cz = d$	Plane in 3D space.
$4, (x, y, z, w)$	$ax + by + cz + dw = e$	Hyperplane

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Linear Equations with More Than Three Variables $(x_1, x_2, x_3, \ldots)$

In **4D space and beyond**, each linear equation represents a **hyperplane** (a higher-dimensional version of a plane that we cannot visualize directly).

Hyperplanes are widely used in **machine learning (SVMs), optimization, and data classification**.

Solution of a Linear Equation

$$\boldsymbol{a_1x_1 + a_2x_2 + a_3x_3 + \cdots + a_nx_n = b} \quad \rightarrow \textbf{(I)}$$

A **solution** to Linear Equation (I) is a sequence of n numbers $s_1, s_2, \ldots, s_n$ which has the property that (I) is satisfied when $x_1 = s_1, x_2 = s_2, \ldots, x_n = s_n$ are substituted in (1).

For example: The equation

$$\boxed{6x_1 - 3x_2 + 4x_3 = -13}$$

has the solution

$$x_1 = 2, \qquad x_2 = 3, \qquad x_3 = -4$$

Because on substituting these values, equation becomes identity i.e.

$$\boxed{6(2) - 3(3) + 4(-4) = -13}$$

$$\boxed{-13 = -13}$$

System of a Linear Equations

A finite set of linear equations is called a system of linear equations or, more briefly, a linear system. The variables are called unknowns.

Linear System in two unknown

$$\boxed{x + 2y = 10}$$
$$\boxed{2x + 3y = 20}$$

Linear System in three unknown

$$\boxed{x + y + 2z = 9}$$
$$\boxed{2x + 4y - 3z = 1}$$
$$\boxed{3x + 6y - 5z = 0}$$

Homogeneous Linear System

A **homogeneous system** has all constant terms **equal to zero**

$$\boxed{3x + 2y + 2z = 0}$$

$$\boxed{2x + 4y - 3z = 0}$$

$$\boxed{3x + 6y - 5z = 0}$$

The system **always** has the **trivial solution**

(0,0,0)

Non-Homogeneous Linear System

A **non-homogeneous system** has at least one **nonzero constant** on the right-hand side

$$\boxed{2x + 3y - z = 5}$$

$$\boxed{x - 4y + 3z = 6}$$

$$\boxed{3x - y + 2z = 8}$$

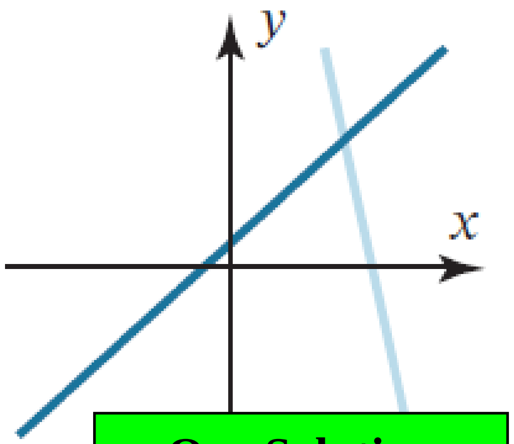
Consistent and Inconsistent Linear System

If the linear system has no solution, it is said to be **inconsistent**, if it has a solution. it is called **consistent.**

A consistent linear system of two equations in two unknowns has either **one solution** or **infinitely many solutions--**there are no other possibilities.

Linear System with Two Variables

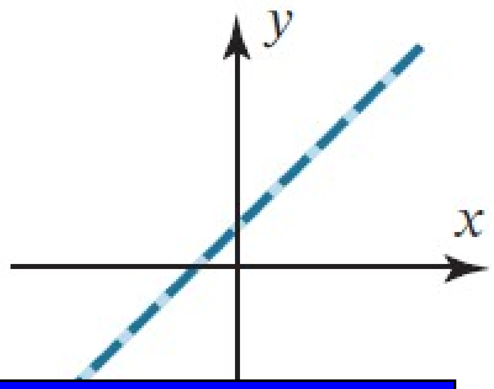
$$x + 2y = 10$$
$$2x + 3y = 20$$



One Solution

(Intersecting lines)

$$x + 2y = 10$$
$$2x + 4y = 20$$

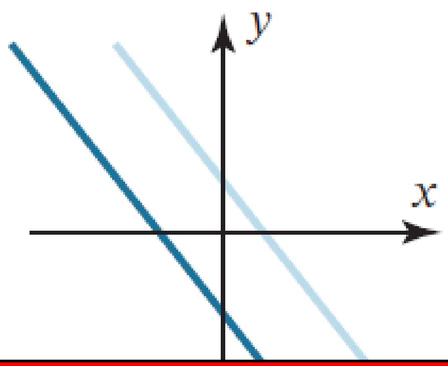


Infinitely many solutions

(Coincident lines)

$$\boxed{0 = 0}$$

$$x + 2y = 10$$
$$2x + 4y = 18$$



No Solution

(Parallel lines)

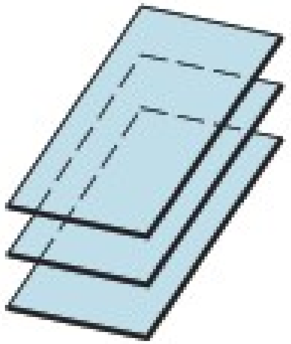
$$\boxed{0 = \text{constant}}$$

Inconsistent System

Consistent System

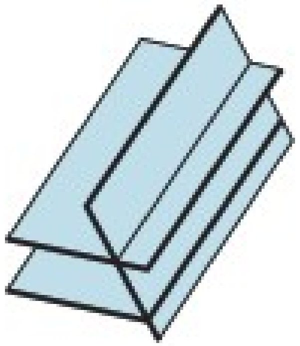
Linear System (inconsistent) in three variables

$$2x - 3y + 4z = 10$$
$$2x - 3y + 4z = 5$$
$$2x - 3y + 4z = 8$$



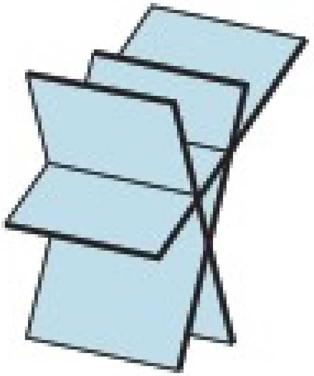
No solutions
(three parallel planes;
no common intersection)

$$2x - 3y + 4z = 10$$
$$2x - 3y + 4z = 5$$
$$2x - 5y + 2z = 11$$



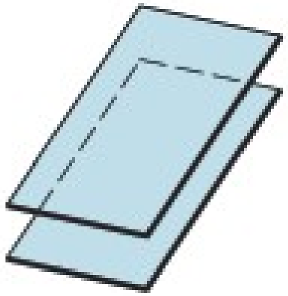
No solutions
(two parallel planes;
no common intersection)

$$x + y + z = 1$$
$$2x - y + 3z = 5$$
$$3x + 4y - z = 0$$



No solutions
(no common intersection)

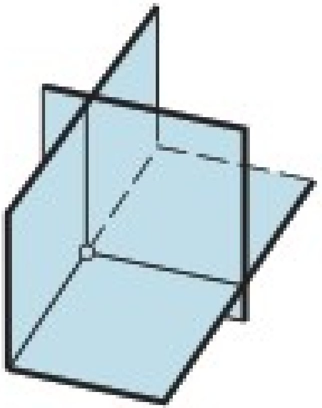
$$2x - 3y + 4z = 5$$
$$4x - 6y + 8z = 10$$
$$2x - 3y + 4z = 7$$



No solutions
(two coincident planes
parallel to the third;
no common intersection)

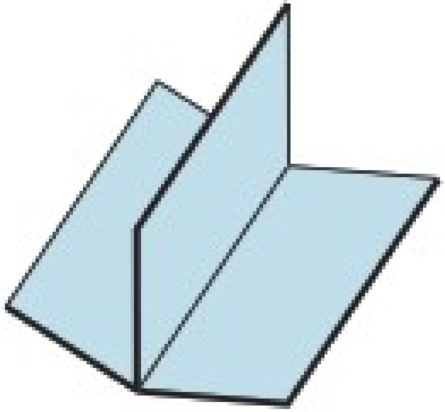
Linear System (consistent) in three variables

$$\begin{aligned} x + y + z &= 6 \\ 2x - y + 3z &= 14 \\ 3x + 4y - 2z &= 4 \end{aligned}$$



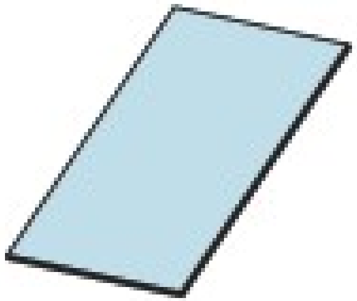
One solution
(intersection is a point)

$$\begin{aligned} x + y + z &= 3 \\ 2\,x - y + z &= 4 \\ 3x + 0y + 2z &= 7 \end{aligned}$$



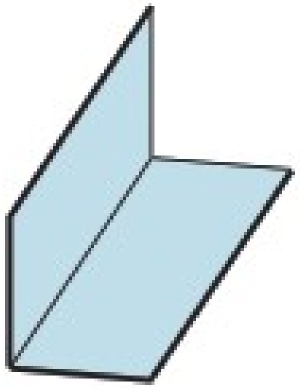
Infinitely many solutions
(intersection is a line)

$$\begin{aligned} x + y + z &= 1 \\ 2x + 2y + 2z &= 2 \\ 3x + 4y + 3z &= 3 \end{aligned}$$



Infinitely many solutions
(planes are all coincident;
intersection is a plane)

$$\begin{aligned} x + y + z &= 1 \\ 2x + 2y + 2z &= 2 \\ x - y + z &= 3 \end{aligned}$$



Infinitely many solutions
(two coincident planes;
intersection is a line)